

TABLE OF CONTENTS

1	INTRODUCTION.....	21
1.1	STUDY OBJECTIVES.....	21
1.2	MARKET DEFINITION.....	21
1.2.1	INCLUSIONS AND EXCLUSIONS	22
1.3	MARKET SCOPE.....	22
1.3.1	MARKETS COVERED	22
1.3.2	YEARS CONSIDERED	23
1.4	CURRENCY.....	23
1.5	LIMITATIONS	23
1.6	STAKEHOLDERS	24
1.7	SUMMARY OF CHANGES.....	24
1.7.1	RECESSION IMPACT	25
2	RESEARCH METHODOLOGY.....	26
2.1	RESEARCH DATA	26
2.1.1	SECONDARY DATA.....	27
2.1.2	PRIMARY DATA.....	28
2.2	MARKET ESTIMATION METHODOLOGY.....	29
2.2.1	INSIGHTS FROM PRIMARY EXPERTS	32
2.3	MARKET GROWTH RATE PROJECTIONS	33
2.4	MARKET BREAKDOWN AND DATA TRIANGULATION	35
2.5	RESEARCH ASSUMPTIONS	36
2.6	RISK ANALYSIS.....	36
2.7	IMPACT OF RECESSION ON MICROBIOME SEQUENCING SERVICES MARKET	36
3	EXECUTIVE SUMMARY.....	38
4	PREMIUM INSIGHTS	42
4.1	MICROBIOME SEQUENCING SERVICES MARKET OVERVIEW	42
4.2	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE AND COUNTRY (2023).....	43
4.3	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)	43

5	MARKET OVERVIEW.....	44
5.1	INTRODUCTION	44
5.2	MARKET DYNAMICS.....	44
5.2.1	DRIVERS	45
5.2.1.1	Increasing demand for metagenomic sequencing.....	45
5.2.1.2	Growing focus on human microbiome therapeutics development.....	46
5.2.1.3	High cost of advanced infrastructure for sequencing.....	47
5.2.2	RESTRAINTS.....	48
5.2.2.1	Dearth of skilled personnel	48
5.2.3	OPPORTUNITIES.....	48
5.2.3.1	Expanding applications of microbiome sequencing in therapeutics development.....	48
5.2.4	CHALLENGES	49
5.2.4.1	Issues related to sample preparation/isolation of microbes for sequencing	49
5.3	TECHNOLOGY ANALYSIS	49
5.4	TRENDS AND DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES	51
5.5	VALUE CHAIN ANALYSIS	52
5.6	ECOSYSTEM MARKET MAP.....	53
5.7	PORTER'S FIVE FORCES ANALYSIS	53
5.7.1	THREAT OF NEW ENTRANTS	53
5.7.2	THREAT OF SUBSTITUTES.....	53
5.7.3	BARGAINING POWER OF SUPPLIERS	53
5.7.4	BARGAINING POWER OF BUYERS	54
5.7.5	INTENSITY OF COMPETITIVE RIVALRY	54
5.8	REGULATORY ANALYSIS.....	54
5.8.1	NORTH AMERICA.....	54
5.8.1.1	US	54
5.8.1.2	Canada.....	55
5.8.2	EUROPE.....	56
5.8.3	ASIA PACIFIC	57
5.8.3.1	China	57
5.8.3.2	Japan.....	57
5.8.3.3	India	58
5.8.4	LATIN AMERICA.....	58
5.8.5	MIDDLE EAST	58
5.8.6	REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS	59
5.9	PRICING ANALYSIS.....	60
5.9.1	AVERAGE SELLING PRICE TREND ANALYSIS	61
5.10	KEY CONFERENCES AND EVENTS IN 2023–2024	62
5.11	KEY STAKEHOLDERS AND BUYING CRITERIA.....	64

6	MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE	65
6.1	INTRODUCTION	66
6.2	SAMPLE PREPARATION	66
6.2.1	NEED FOR SPECIFIC SAMPLE PREPARATION PROTOCOLS BASED ON SAMPLE TYPE TO DRIVE GROWTH	66
6.3	SEQUENCING AND LIBRARY PREPARATION	68
6.3.1	SEQUENCING AND LIBRARY PREPARATION SERVICES TO DOMINATE MARKET DURING FORECAST PERIOD.....	68
6.4	DATA ANALYSIS	70
6.4.1	DEVELOPMENT OF ADVANCED BIOINFORMATICS SOLUTIONS TO DRIVE GROWTH	70
7	MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE	73
7.1	INTRODUCTION	74
7.2	AMPLICON SEQUENCING.....	74
7.2.1	HIGH ACCURACY OF 16S AMPLICON SEQUENCING TO DRIVE ADOPTION	74
7.3	WHOLE-GENOME SEQUENCING	76
7.3.1	ACCURATE REFERENCE GENOMES GENERATED FOR MICROBIAL IDENTIFICATION TO BOOST MARKET.....	76
7.4	SHOTGUN SEQUENCING	78
7.4.1	ADVANCEMENTS IN DATA ANALYSIS SOLUTIONS TO SUPPORT ADOPTION.....	78
7.5	OTHER TYPES	81
8	MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY	83
8.1	INTRODUCTION	84
8.2	SEQUENCING BY SYNTHESIS (SBS).....	84
8.2.1	SBS TECHNOLOGY ENABLES HIGHEST PRODUCTION OF BASE PAIRS IN SHORT TIME	84
8.3	SEQUENCING BY LIGATION (SBL)	86
8.3.1	INCREASING DEMAND FOR SBL IN FOOD MICROBIOME RESEARCH TO BOOST GROWTH.....	86
8.4	NANOPORE SEQUENCING.....	88
8.4.1	ONLY SEQUENCING TECHNOLOGY WITH DIRECT RNA SEQUENCING CAPABILITIES.....	88
8.5	OTHER TECHNOLOGIES.....	91
9	MICROBIOME SEQUENCING SERVICES MARKET, BY END USER	93
9.1	INTRODUCTION	94
9.2	PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES	94
9.2.1	ROBUST FUNDING FOR MICROBIOME-BASED THERAPEUTICS TO DRIVE MARKET	94
9.3	ACADEMIC AND RESEARCH INSTITUTES.....	97
9.3.1	EXPANDING APPLICATIONS OF MICROBIOME RESEARCH TO DRIVE GROWTH	97
9.4	OTHER END USERS.....	99

10	MICROBIOME SEQUENCING SERVICES MARKET, BY REGION	102
10.1	INTRODUCTION	103
10.2	NORTH AMERICA	104
10.2.1	US	107
10.2.1.1	Increasing awareness of microbiome research to drive growth	107
10.2.2	CANADA.....	109
10.2.2.1	Increasing prevalence of chronic diseases to boost microbiome research.....	109
10.2.3	NORTH AMERICA: IMPACT OF RECESSION	111
10.3	EUROPE	112
10.3.1	GERMANY	114
10.3.1.1	Rapidly increasing government funding to support growth	114
10.3.2	UK	116
10.3.2.1	Significant government investments in genomics research to propel growth.....	116
10.3.3	FRANCE	118
10.3.3.1	Growing number of microbiome sequencing startups to propel growth	118
10.3.4	ITALY.....	121
10.3.4.1	Increasing use of microbiome services in clinical research to drive growth.....	121
10.3.5	SPAIN	123
10.3.5.1	Increasing prevalence of chronic diseases to drive market	123
10.3.6	REST OF EUROPE	124
10.3.7	EUROPE: IMPACT OF RECESSION.....	126
10.4	ASIA PACIFIC.....	127
10.4.1	CHINA.....	130
10.4.1.1	Increasing investments in R&D to drive growth	130
10.4.2	JAPAN	132
10.4.2.1	Significant growth in geriatric population to present opportunities for market growth	132
10.4.3	INDIA	134
10.4.3.1	Increasing prevalence of chronic diseases to support growth	134
10.4.4	REST OF ASIA PACIFIC.....	136
10.4.5	ASIA PACIFIC: IMPACT OF RECESSION	138
10.5	LATIN AMERICA	138
10.5.1	BRAZIL.....	141
10.5.1.1	Growing adoption of next-generation sequencing technologies to propel growth.....	141
10.5.2	REST OF LATIN AMERICA	142
10.5.3	LATIN AMERICA: IMPACT OF RECESSION	144
10.6	MIDDLE EAST & AFRICA.....	145
10.6.1	MIDDLE EAST & AFRICA: IMPACT OF RECESSION	146

11	COMPETITIVE LANDSCAPE	147
11.1	OVERVIEW	147
11.2	STRATEGIES ADOPTED BY KEY PLAYERS	148
11.3	MARKET SHARE ANALYSIS	149
11.4	REVENUE SHARE ANALYSIS OF TOP MARKET PLAYERS	150
11.5	COMPANY EVALUATION QUADRANT	151
11.5.1	STARS.....	151
11.5.2	EMERGING LEADERS.....	151
11.5.3	PERVASIVE PLAYERS	151
11.5.4	PARTICIPANTS	151
11.6	COMPETITIVE SCENARIO	153
12	COMPANY PROFILES.....	155
12.1	KEY PLAYERS.....	155
12.1.1	CHARLES RIVER LABORATORIES.....	155
12.1.1.1	Business overview	155
12.1.1.2	Services offered	156
12.1.1.3	Recent developments	157
12.1.1.4	MnM view.....	157
12.1.1.4.1	Key strengths	157
12.1.1.4.2	Strategic choices	157
12.1.1.4.3	Weaknesses and competitive threats	157
12.1.2	EUROFINS SCIENTIFIC.....	158
12.1.2.1	Business overview	158
12.1.2.2	Services offered	159
12.1.2.3	MnM view.....	159
12.1.2.3.1	Key strengths	159
12.1.2.3.2	Strategic choices	159
12.1.2.3.3	Weaknesses and competitive threats	159
12.1.3	BGI GROUP	160
12.1.3.1	Business overview	160
12.1.3.2	Services offered	160
12.1.4	COSMOSID	161
12.1.4.1	Business overview	161
12.1.4.2	Services offered	161
12.1.4.3	Recent developments	162
12.1.5	MICROBA.....	163
12.1.5.1	Business overview	163
12.1.5.2	Services offered	163
12.1.5.3	Recent developments	163

12.1.6	QIAGEN N.V.....	165
12.1.6.1	Business overview	165
12.1.6.2	Services offered	166
12.1.6.3	Recent developments	167
12.1.7	MICROBIOME INSIGHTS.....	168
12.1.7.1	Business overview	168
12.1.7.2	Services offered	168
12.1.7.3	Recent developments	169
12.1.8	BASECLEAR	170
12.1.8.1	Business overview	170
12.1.8.2	Services offered	170
12.1.9	CD GENOMICS	171
12.1.9.1	Business overview	171
12.1.9.2	Services offered	171
12.1.10	ZYMO RESEARCH CORPORATION.....	173
12.1.10.1	Business overview	173
12.1.10.2	Services offered	173
12.1.11	ORASURE TECHNOLOGIES.....	175
12.1.11.1	Business overview	175
12.1.11.2	Services offered	176
12.1.11.3	Recent developments	177
12.1.12	MR DNA.....	178
12.1.12.1	Business overview	178
12.1.12.2	Services offered	178
12.1.13	EREMID GENOMIC SERVICES	179
12.1.13.1	Business overview	179
12.1.13.2	Services offered	179
12.1.14	CLINICAL-MICROBIOMICS A/S.....	180
12.1.14.1	Business overview	180
12.1.14.2	Services offered	180
12.1.14.3	Recent developments	181
12.1.15	NOVOGENE CO., LTD.....	182
12.1.15.1	Business overview	182
12.1.15.2	Services offered	182
12.2	OTHER PLAYERS	183
12.2.1	EZBIOME	183
12.2.2	BOSTER BIOLOGICAL TECHNOLOGY	183
12.2.3	ZIFO.....	184
12.2.4	OMICS2VIEW.CONSULTING GBR	184
12.2.5	MACROGEN, INC.	185

13	APPENDIX.....	186
13.1	DISCUSSION GUIDE	186
13.2	KNOWLEDGESTORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL	189
13.3	CUSTOMIZATION OPTIONS	191
13.4	RELATED REPORTS	191
13.5	AUTHOR DETAILS.....	192

LIST OF TABLES

TABLE 1	GLOBAL INFLATION RATE PROJECTIONS, 2021–2027 (% GROWTH)	37
TABLE 2	US HEALTH EXPENDITURE, 2019–2022 (USD MILLION)	37
TABLE 3	US HEALTH EXPENDITURE, 2023–2027 (USD MILLION)	37
TABLE 4	MICROBIOME SEQUENCING SERVICES MARKET: IMPACT ANALYSIS OF DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES	45
TABLE 5	AVERAGE SELLING PRICE OF NGS SYSTEMS, BY KEY PLAYER (2021)	47
TABLE 6	MICROBIOME SEQUENCING SERVICES MARKET: PORTER'S FIVE FORCES ANALYSIS	53
TABLE 7	US FDA: MEDICAL DEVICE CLASSIFICATION	55
TABLE 8	US: MEDICAL DEVICE REGULATORY APPROVAL PROCESS	55
TABLE 9	CHINA: CLASSIFICATION OF MEDICAL DEVICES	57
TABLE 10	REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS: GENOMICS, MICROBIOME RESEARCH, AND METAGENOMICS	59
TABLE 11	AVERAGE SELLING PRICE, BY SERVICE AND TYPE	60
TABLE 12	MICROBIOME SEQUENCING/METAGENOMIC SEQUENCING/NGS CONFERENCES, 2023–2024	62
TABLE 13	MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	66
TABLE 14	MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY REGION, 2021–2028 (USD MILLION)	67
TABLE 15	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	67
TABLE 16	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	67
TABLE 17	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	68
TABLE 18	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	68
TABLE 19	MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY REGION, 2021–2028 (USD MILLION)	69
TABLE 20	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	69
TABLE 21	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	69
TABLE 22	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	70
TABLE 23	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)	70
TABLE 24	MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY REGION, 2021–2028 (USD MILLION)	71
TABLE 25	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)	71
TABLE 26	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)	71
TABLE 27	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)	72

TABLE 28	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)	72
TABLE 29	MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	74
TABLE 30	AMPLICON SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)	75
TABLE 31	NORTH AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	75
TABLE 32	EUROPE: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	75
TABLE 33	ASIA PACIFIC: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	76
TABLE 34	LATIN AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	76
TABLE 35	WHOLE-GENOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)	77
TABLE 36	NORTH AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	77
TABLE 37	EUROPE: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	77
TABLE 38	ASIA PACIFIC: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	78
TABLE 39	LATIN AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	78
TABLE 40	SHOTGUN SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)	79
TABLE 41	NORTH AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	79
TABLE 42	EUROPE: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	80
TABLE 43	ASIA PACIFIC: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	80
TABLE 44	LATIN AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	80
TABLE 45	OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)	81
TABLE 46	NORTH AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	81
TABLE 47	EUROPE: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	82
TABLE 48	ASIA PACIFIC: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	82
TABLE 49	LATIN AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	82
TABLE 50	MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	84
TABLE 51	MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY REGION, 2021–2028 (USD MILLION)	85
TABLE 52	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)	85
TABLE 53	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)	85

TABLE 54	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)	86
TABLE 55	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)	86
TABLE 56	MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY REGION, 2021–2028 (USD MILLION)	87
TABLE 57	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)	87
TABLE 58	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)	87
TABLE 59	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)	88
TABLE 60	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)	88
TABLE 61	MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY REGION, 2021–2028 (USD MILLION)	89
TABLE 62	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)	89
TABLE 63	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)	90
TABLE 64	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)	90
TABLE 65	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)	90
TABLE 66	MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY REGION, 2021–2028 (USD MILLION)	91
TABLE 67	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)	91
TABLE 68	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)	92
TABLE 69	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)	92
TABLE 70	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)	92
TABLE 71	MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	94
TABLE 72	MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY REGION, 2021–2028 (USD MILLION)	95
TABLE 73	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)	95
TABLE 74	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)	96
TABLE 75	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)	96
TABLE 76	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)	96
TABLE 77	MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY REGION, 2021–2028 (USD MILLION)	97

TABLE 78	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)	98
TABLE 79	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)	98
TABLE 80	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)	98
TABLE 81	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)	99
TABLE 82	MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY REGION, 2021–2028 (USD MILLION)	100
TABLE 83	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)	100
TABLE 84	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)	100
TABLE 85	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)	101
TABLE 86	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)	101
TABLE 87	MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)	104
TABLE 88	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	106
TABLE 89	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	106
TABLE 90	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	106
TABLE 91	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	107
TABLE 92	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	107
TABLE 93	US: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	108
TABLE 94	US: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	108
TABLE 95	US: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	109
TABLE 96	US: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	109
TABLE 97	CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	110
TABLE 98	CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	110
TABLE 99	CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	111
TABLE 100	CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	111
TABLE 101	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	113
TABLE 102	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	113
TABLE 103	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	113

TABLE 104	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	114
TABLE 105	EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	114
TABLE 106	GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	115
TABLE 107	GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	115
TABLE 108	GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	116
TABLE 109	GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	116
TABLE 110	UK: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	117
TABLE 111	UK: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	117
TABLE 112	UK: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	118
TABLE 113	UK: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	118
TABLE 114	HUMAN MICROBIOME STARTUPS IN FRANCE	119
TABLE 115	FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	119
TABLE 116	FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	120
TABLE 117	FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	120
TABLE 118	FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	120
TABLE 119	ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	121
TABLE 120	ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	122
TABLE 121	ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	122
TABLE 122	ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	122
TABLE 123	SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	123
TABLE 124	SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	123
TABLE 125	SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	124
TABLE 126	SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	124
TABLE 127	REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	125
TABLE 128	REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	125
TABLE 129	REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	125

TABLE 130	REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	126
TABLE 131	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	129
TABLE 132	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	129
TABLE 133	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	129
TABLE 134	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	130
TABLE 135	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	130
TABLE 136	CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	131
TABLE 137	CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	131
TABLE 138	CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	132
TABLE 139	CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	132
TABLE 140	JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	133
TABLE 141	JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	133
TABLE 142	JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	134
TABLE 143	JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	134
TABLE 144	INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	135
TABLE 145	INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	135
TABLE 146	INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	136
TABLE 147	INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	136
TABLE 148	REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	137
TABLE 149	REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	137
TABLE 150	REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	137
TABLE 151	REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	138
TABLE 152	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)	139
TABLE 153	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	139
TABLE 154	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	139
TABLE 155	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	140

TABLE 156	LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	140
TABLE 157	BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	141
TABLE 158	BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	141
TABLE 159	BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	142
TABLE 160	BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	142
TABLE 161	REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	143
TABLE 162	REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	143
TABLE 163	REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	143
TABLE 164	REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	144
TABLE 165	MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)	145
TABLE 166	MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)	145
TABLE 167	MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)	146
TABLE 168	MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)	146
TABLE 169	MICROBIOME SEQUENCING SERVICES MARKET: DEGREE OF COMPETITION	149
TABLE 170	SERVICE LAUNCHES	153
TABLE 171	DEALS	153
TABLE 172	CHARLES RIVER LABORATORIES: BUSINESS OVERVIEW	155
TABLE 173	EUROFINS SCIENTIFIC: BUSINESS OVERVIEW	158
TABLE 174	BGI GROUP: BUSINESS OVERVIEW	160
TABLE 175	COSMOSID: BUSINESS OVERVIEW	161
TABLE 176	MICROBA: BUSINESS OVERVIEW	163
TABLE 177	QIAGEN N.V.: BUSINESS OVERVIEW	165
TABLE 178	MICROBIOME INSIGHTS: BUSINESS OVERVIEW	168
TABLE 179	BASECLEAR: BUSINESS OVERVIEW	170
TABLE 180	CD GENOMICS: COMPANY OVERVIEW	171
TABLE 181	ZYMO RESEARCH CORPORATION: COMPANY OVERVIEW	173
TABLE 182	ORASURE TECHNOLOGIES: BUSINESS OVERVIEW	175
TABLE 183	MR DNA: BUSINESS OVERVIEW	178
TABLE 184	EREMID GENOMIC SERVICES: BUSINESS OVERVIEW	179
TABLE 185	CLINICAL-MICROBIOMICS A/S: BUSINESS OVERVIEW	180
TABLE 186	NOVOGENE CO., LTD.: BUSINESS OVERVIEW	182

LIST OF FIGURES

FIGURE 1	RESEARCH DESIGN	26
FIGURE 2	MICROBIOME SEQUENCING SERVICES MARKET: BREAKDOWN OF PRIMARIES	28
FIGURE 3	MICROBIOME SEQUENCING SERVICES MARKET SIZE ESTIMATION (SUPPLY-SIDE ANALYSIS), 2022	29
FIGURE 4	MARKET SIZE ESTIMATION: APPROACH 1 (COMPANY REVENUE ANALYSIS-BASED ESTIMATION), 2022	30
FIGURE 5	ILLUSTRATIVE EXAMPLE OF COMPANY REVENUE ANALYSIS, 2022	31
FIGURE 6	MARKET SIZE VALIDATION FROM PRIMARY SOURCES	32
FIGURE 7	MICROBIOME SEQUENCING SERVICES MARKET (SUPPLY-SIDE): CAGR PROJECTIONS	33
FIGURE 8	MICROBIOME SEQUENCING SERVICES MARKET: GROWTH ANALYSIS OF DEMAND-SIDE DRIVERS	34
FIGURE 9	DATA TRIANGULATION METHODOLOGY	35
FIGURE 10	MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2023 VS. 2028 (USD MILLION)	38
FIGURE 11	MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)	39
FIGURE 12	MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2023 VS. 2028 (USD MILLION)	39
FIGURE 13	MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2023 VS. 2028 (USD MILLION)	40
FIGURE 14	GEOGRAPHICAL SNAPSHOT OF MICROBIOME SEQUENCING SERVICES MARKET	41
FIGURE 15	GROWING FOCUS ON HUMAN MICROBIOME THERAPEUTICS DEVELOPMENT TO DRIVE MARKET	42
FIGURE 16	SEQUENCING AND LIBRARY PREPARATION SERVICES TO ACCOUNT FOR LARGEST SHARE OF NORTH AMERICAN MICROBIOME SEQUENCING SERVICES MARKET IN 2023	43
FIGURE 17	SHOTGUN SEQUENCING SEGMENT TO GROW AT HIGHEST CAGR DURING FORECAST PERIOD	43
FIGURE 18	MICROBIOME SEQUENCING SERVICES MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES	44
FIGURE 19	ROLE OF MICROBIOME SEQUENCING IN THERAPEUTICS DEVELOPMENT	46
FIGURE 20	REVENUE SHIFT AND NEW REVENUE POCKETS FOR MICROBIOME SEQUENCING SERVICE PROVIDERS	51
FIGURE 21	VALUE CHAIN ANALYSIS OF MICROBIOME SEQUENCING SERVICES MARKET	52
FIGURE 22	ECOSYSTEM OF MICROBIOME SEQUENCING SERVICES MARKET	53
FIGURE 23	INFLUENCE OF STAKEHOLDERS ON BUYING PROCESS OF SEQUENCING SERVICES	64
FIGURE 24	CRITERIA FOR OUTSOURCING SEQUENCING SERVICES	64
FIGURE 25	MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2023 VS. 2028 (USD MILLION)	103
FIGURE 26	NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT	105
FIGURE 27	ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT	128
FIGURE 28	STRATEGIES ADOPTED BY KEY PLAYERS IN MICROBIOME SEQUENCING SERVICES MARKET, 2020–2023	148
FIGURE 29	MICROBIOME SEQUENCING SERVICES MARKET SHARE ANALYSIS, BY KEY PLAYER, 2022	149